

TAK Server Certificate Enrollment/Revoking

Certificate Enrollment - Hard vs Soft

- With the makeCert.sh we are creating "Soft" certificates which are manually generated followed by the UserManager.jar placing the certificate in a group.
 - You would need to pass the user both the trust store certificate and the user certificate for them to be able to connect.
- The other method is "Hard", using certificate enrollment where the client gets its certificate directly from the server and is placed in the group automatically.
 - By selecting the auto enrollment feature when connecting to the server you only need to enter in the IP address and then enter the credentials.
 - Unless you don't have a public facing domain, then you will need to pass the user the trust store cert.

Certificate Auto Enrollment

- Identify our signing truststore
 - **ls -l /opt/tak/certs/files/*-signing.jks**
- Navigate to your TAK Server admin dashboard on a browser
 - Navigate to: Hamburger menu → Configuration → Security and Authentication
 - Under the Security Configuration section select the “Edit Security” option.
 - Enable the “Enable Certificate Enrollment” option and then enable the “TAK Server CA” option.
 - Fill in the following:
 - Signing Keystore File = certs/files/<CACommonName>-signing.jks
 - the common name if the one we identified as our signing truststore above.
 - Signing Keystore Password = Default: atakatak (if you changed the password in the cert-metadata.sh file that would go here.)
 - Validity Days: Leave as default or change to how many days the certificate should be valid for.
 - After all those are filled in click “submit”
 - Restart the server:
 - **sudo systemctl restart takserver**

Certificate Auto Enrollment - Command Line

- To change the name of the server values:
 - Enter the “tak” user
 - **sudo su tak**
 - Change the directory to the tak directory
 - **cd /opt/tak**
 - Edit the CoreConfig.xml file
 - **vi CoreConfig.xml**
 - Scroll down to find the following:
 - modify the “TAK” value to your desired output.
 - O = Organization
 - OU = Organization Unit

```
certificateSigning CA= TAKSERVER /
<certificateConfig>
  <nameEntries>
    <nameEntry name="O" value="ORG"/>
    <nameEntry name="OU" value="TAK"/>
    <nameEntry name="OU" value="Users"/>
  </nameEntries>
</certificateConfig>
<TAKServerCAConfig keyStore="JKS" keyStoreFile="cert
```

Certificate Auto Enrollment - Command Line

- In the same text editor, we need to add CAkey and CAcertificate to the TAKServerCAConfig line to support revocations.
 - `<TAKServerCAConfig keystore="JKS" keystoreFile="certs/files/TAK-ID-CA-01-signing.jks" keystorePass="atakatak" validityDays="30" signatureAlg="SHA256WithRSA" CAkey="/opt/tak/certs/files/<CAcommonName>" CAcertificate="/opt/tak/certs/files/<CAcommonName>"/>`
 - **Example:** `<TAKServerCAConfig keystore="JKS" keystoreFile="certs/files/TAK-ID-CA-01-signing.jks" keystorePass="atakatak" validityDays="30" signatureAlg="SHA256WithRSA" CAkey="/opt/tak/certs/files/TAK-ID-CA-01" CAcertificate="/opt/tak/certs/files/TAK-ID-CA-01"/>`

```
</nameCertificate>
</certificateConfig>
<TAKServerCAConfig keystore="JKS" keystoreFile="certs/files/TAK-ID-CA-01-signing.jks" keystorePass="atakatak" validityDays="30" signatureAlg="SHA256WithRSA" CAkey="/opt/tak/certs/files/TAK-ID-CA-01" CAcertificate="/opt/tak/certs/files/TAK-ID-CA-01"/>
</certificateSigning>
```

```
</certificateConfig>
<TAKServerCAConfig keystore="JKS" keystoreFile="certs/files/test-CA-signing.jks" keystorePass="password" validityDays="30" signatureAlg="SHA256WithRSA" CAkey="/opt/tak/certs/files/test-CA" CAcertificate="/opt/tak/certs/files/test-CA" />
```

Certificate Auto Enrollment

- Lastly we need to add the following line to our `<auth>` element:
 - **`x509checkRevocation="true"`**
- Save and Exit
- To verify our changes run the following:
 - **`./validateConfig.sh`**
- Exit the tak user
 - **`exit`**
- Restart the `takserver.service`
 - **`sudo systemctl restart takserver.service`**
- Monitor the logs for errors
 - **`tail -f /opt/tak/logs/takserver-messaging.log`**

```
<announce/>
</network>
<auth x509groups="true" x509addAnonymous="false" x509checkRevocation="true">
  <File location="UserAuthenticationFile.xml"/>
</auth>
```

```
bash-4.4$ ./validateConfig.sh
CoreConfig.xml validates
bash-4.4$ vi CoreConfig.xml
bash-4.4$
```

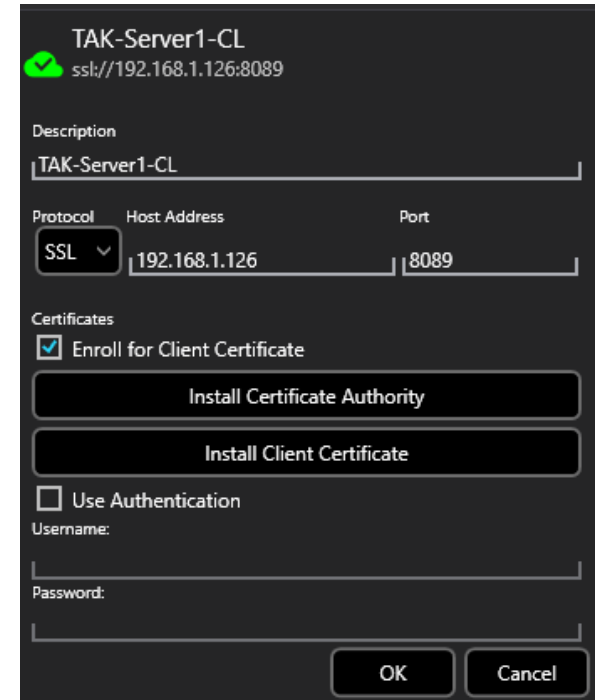
Enrolling Clients with Auto Enrollment

- In the admin dashboard open the following: Hamburger menu → Administrative → Manage Users
- Click on “Add User”
 - fill in the details and select “Create New User”
- The certificate enrollment process is communicated to the server over TCP port 8446, which is not enabled by default.
 - Check if the port is already enabled:
 - **sudo firewall-cmd --list-all**
 - Enable 8446 Certificate enrollment port.
 - **sudo firewall-cmd --zone=public --add-port 8446/tcp --permanent**
 - Reload the firewall to apply the changes
 - **sudo firewall-cmd --reload**

```
[takadmin@localhost ~]$ sudo firewall-cmd --zone=public --add-port 8446/tcp --permanent
[sudo] password for takadmin:
success
[takadmin@localhost ~]$ sudo firewall-cmd --reload
success
[takadmin@localhost ~]$
```

Connecting ATAK to TAK Server

- Select the hamburger menu in the top right
- Scroll down and select settings
- Select “Network Preferences”
- Select “Tak Servers”
- In the top right select the three dots and then “Add”
 - Name:<name you like for the server>
 - Address:<IP address>
 - Enable “Enroll for Client Certificate”
 - Select “Ok”
 - Note: If you are trying to connect to the TAK Server without setting up the public facing domain you need to pass the truststore cert as well or create a data package.
- Enter in the user credentials that you create from the admin dashboard



The screenshot shows a configuration window titled "TAK-Server1-CL" with a green checkmark icon and the URL "ssl://192.168.1.126:8089". The "Description" field contains "TAK-Server1-CL". Below this is a table with three columns: "Protocol", "Host Address", and "Port". The "Protocol" is set to "SSL", "Host Address" is "192.168.1.126", and "Port" is "8089". Under the "Certificates" section, the checkbox "Enroll for Client Certificate" is checked. There are two buttons: "Install Certificate Authority" and "Install Client Certificate". The "Use Authentication" checkbox is unchecked. Below it are fields for "Username:" and "Password:". At the bottom right are "OK" and "Cancel" buttons.

Protocol	Host Address	Port
SSL	192.168.1.126	8089

Certificates

☒ Enroll for Client Certificate

☐ Use Authentication

Username:

Password:

OK Cancel

Soft Certificate Management

- Switch to the “tak” user
 - **sudo su tak**
- navigate to the cert's directory
 - **cd /opt/tak/certs**
- Create client certificate
 - **./makeCert.sh client <ClientName>**
 - Example: `./makeCert.sh client wintak`
- Add the client to a group
 - **java -jar /opt/tak/utils/UserManager.jar certmod -g __ANON__ /opt/tak/certs/files/<commonName>**
 - **Example:** `java -jar /opt/tak/utils/UserManager.jar certmod -g __ANON__ /opt/tak/certs/files/wintak.pem`
 - We need to place them in a group or else they will not be available in the dashboard.
 - If you would like to create a new group for the user, you can replace “__ANON__” with the new group

Soft Certificate Management

- Copy the files to move to our admin workstation
 - Exit the tak user
 - **exit**
 - Change the directory to the home directory
 - **cd ~**
 - Copy our client and public CA certificates
 - **sudo cp -v /opt/tak/certs/files/wintak.p12 .**
 - **sudo cp -v /opt/tak/certs/files/truststore-<public CA name>.p12 .**
 - **Example:** **sudo cp -v /opt/tak/certs/files/truststore-TAK-ID-CA-01.p12 .**
 - Note: if you already have the truststore cert downloaded you do not need to do it again.
 - Restore ownership of the files to the home directory.
 - **sudo chown -R takadmin:takadmin /home/takadmin**

```
bash-4.4$ exit
exit
[takadmin@localhost ~]$ cd ~
[takadmin@localhost ~]$ sudo cp -v /opt/tak/certs/files/wintak.p12 .
[sudo] password for takadmin:
'/opt/tak/certs/files/wintak.p12' -> './wintak.p12'
[takadmin@localhost ~]$ sudo cp -fv /opt/tak/certs/files/truststore-TAK-ID-CA-01.p12 .
'/opt/tak/certs/files/truststore-TAK-ID-CA-01.p12' -> './truststore-TAK-ID-CA-01.p12'
[takadmin@localhost ~]$ sudo chown -R takadmin:takadmin /home/takadmin
[takadmin@localhost ~]$
```

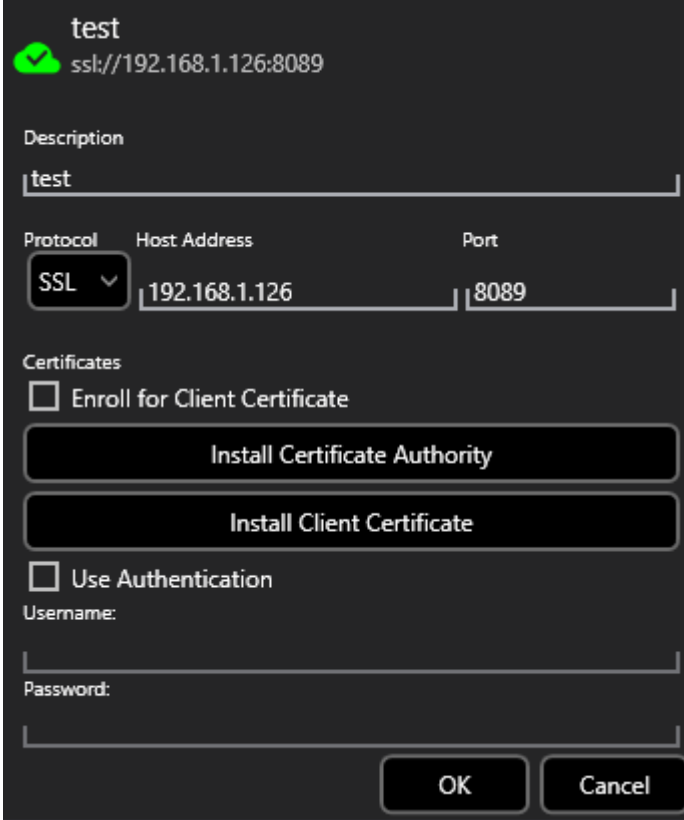
Soft Certificate Management

- Download the files from your Linux machine to your local machine.
- Navigate to the folder location you want to certificates to be stored and open a terminal at this location.
- Copy over the files:
 - **scp <username>@<IP address>:~/<file> .**
 - **Example:** scp [takadmin@192.168.1.104:~/wintak.p12](#) .
 - **Example:** scp [takadmin@192.168.1.104:~/truststore-*.p12](#) .

```
PS C:\Users\huntr\Documents\TAK Server\TAK Servers\TAK-Server-CL\MASTER_SERVER_NAME_FOLDER_STRUCTURE-20240731T121233Z-001\MASTER_SERVER_NAME_FOLDER_STRUCTURE\Cert Files> scp takadmin@192.168.1.104:~/wintak.p12 .
takadmin@192.168.1.104's password:
wintak.p12
100% 4378 1.4MB/s 00:00
PS C:\Users\huntr\Documents\TAK Server\TAK Servers\TAK-Server-CL\MASTER_SERVER_NAME_FOLDER_STRUCTURE-20240731T121233Z-001\MASTER_SERVER_NAME_FOLDER_STRUCTURE\Cert Files> scp takadmin@192.168.1.104:~/truststore-*.p12 .
takadmin@192.168.1.104's password:
truststore-TAK-ID-CA-01.p12
100% 2096 1.0MB/s 00:00
PS C:\Users\huntr\Documents\TAK Server\TAK Servers\TAK-Server-CL\MASTER_SERVER_NAME_FOLDER_STRUCTURE-20240731T121233Z-001\MASTER_SERVER_NAME_FOLDER_STRUCTURE\Cert Files>
```

Soft Certificate Management - Connecting

- To connect to the TAK Server from a TAK device you will need both the client cert and the truststore cert downloaded on the device.
- Once in the TAK devices “Manage Server Connections” section you will select “Add Item”
- You will then fill in all the necessary information and import the certificate authority (truststore cert) and the client cert.
- With both these certs you will not need to authenticate when connecting to the server.



The screenshot shows a dialog box titled "test" with a green checkmark icon. The URL "ssl://192.168.1.126:8089" is displayed. Below the title, there is a "Description" field containing "test". A table with three columns: "Protocol", "Host Address", and "Port" is shown. The "Protocol" is set to "SSL" (with a dropdown arrow), "Host Address" is "192.168.1.126", and "Port" is "8089". Under the "Certificates" section, there is a checkbox for "Enroll for Client Certificate" which is unchecked. Below this are two buttons: "Install Certificate Authority" and "Install Client Certificate". Further down, there is a checkbox for "Use Authentication" which is unchecked, followed by "Username:" and "Password:" labels with corresponding input fields. At the bottom right, there are "OK" and "Cancel" buttons.

Protocol	Host Address	Port
SSL	192.168.1.126	8089

Certificates

☐ Enroll for Client Certificate

Install Certificate Authority

Install Client Certificate

☐ Use Authentication

Username:

Password:

OK Cancel

Revoking Certificates - Server Dashboard

- If the server is configured for certificate autoenrollment we can revoke certificates from the server dashboard.
 - From the server dashboard → hamburger menu → administrative → Client Certificates
 - You can check the certificates you want to revoke then select the “Revoke Selected” option at the top.
 - You can also go the cert you want to revoke and select revoke on the far right of that cert.
 - You will get a warning about the server needing to restart to update the CRL (Certificate Revocation List).

Client Certificates

[Show All](#) [Show Expired](#) [Show Revoked](#) [Show Active](#) [Show Replaced](#)

[Download Selected](#) [Delete Selected](#) [Revoke Selected](#) [Select All](#)

Select	Creator	Subject	User	Device UID	Serial	Issuance Date ▲	Expiration Date	Effective Date	Revocation Date
☐	player2	OU=TESTINGING,OU=TESTING,O=TEST,CN=player2	player2	ANDROID-354267099049014	61c3df6d	2024-08-02T19:48:15.265Z	2024-09-01T19:48:15.255Z	2024-08-02T07:48:15.255Z	
☐	atak	OU=TESTINGING,OU=TESTING,O=TEST,CN=atak	atak	ANDROID-354267099049014	5c4b85b3	2024-08-05T14:18:39.252Z	2024-09-04T14:18:39.244Z	2024-08-05T02:18:39.244Z	2024-08-05T14:41

Revoking Certificates - Server Monitoring

- When the server restarts you can monitor the log file (takserver-messaging.log) to see the server revoking the client.
 - 2024-01-17-20:26:27.758 [epollEventLoopGroup-4-1] ERROR c.b.m.nio.codec.impls.X509AuthCodec - X509 auth exception info: CN: myTeckNet. Message: **Attempt to use revoked certificate** : OU=TAK,O=TAK,CN=myTeckNet com.bbn.marti.remote.exception.RevokedException: Attempt to use revoked certificate : OU=TAK,O=TAK,CN=myTeckNet at com.bbn.marti.groups.X509Authenticator.auth(X509Authenticator.java:134)
...
 - 2024-01-17-20:26:27.760 [epollEventLoopGroup-4-1] ERROR c.b.m.nio.codec.impls.X509AuthCodec - TakException in doTlsAuth Attempt to use revoked certificate : OU=TAK,O=TAK,CN=myTeckNet com.bbn.marti.remote.exception.RevokedException: **Attempt to use revoked certificate** : OU=TAK,O=TAK,CN=myTeckNet at com.bbn.marti.groups.X509Authenticator.auth(X509Authenticator.java:134)
- You will also see the server attempt to add the client then immediately remove it, incrementing the “id=tls” every time.
 - 2024-01-17-20:45:07.106 [epollEventLoopGroup-4-2] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:19 source=192.168.178.177
 - 2024-01-17-20:45:07.108 [takserver-scheduled-1] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:19

```
[takadmin@localhost logs]$ tail -f /opt/tak/logs/takserver-messaging.log
2024-08-05-10:49:53.043 [epollEventLoopGroup-4-4] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:34 source=192.168.1.242
2024-08-05-10:49:53.091 [takserver-scheduled-2] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:34
2024-08-05-10:50:08.686 [epollEventLoopGroup-4-5] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:35 source=192.168.1.242
2024-08-05-10:50:08.709 [takserver-scheduled-2] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:35
2024-08-05-10:50:24.326 [epollEventLoopGroup-4-6] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:36 source=192.168.1.242
2024-08-05-10:50:24.348 [takserver-scheduled-0] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:36
2024-08-05-10:50:39.940 [epollEventLoopGroup-4-1] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:37 source=192.168.1.242
2024-08-05-10:50:39.960 [takserver-scheduled-1] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:37
2024-08-05-10:50:55.082 [epollEventLoopGroup-4-2] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:38 source=192.168.1.242
2024-08-05-10:50:55.103 [takserver-scheduled-1] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:38
2024-08-05-10:51:10.721 [epollEventLoopGroup-4-3] INFO c.b.m.s.DistributedSubscriptionManager - Added Subscription: id=tls:39 source=192.168.1.242
2024-08-05-10:51:10.761 [takserver-scheduled-2] INFO c.b.m.s.DistributedSubscriptionManager - Removed Subscription: tls:39
```

Certificate Revocation List

- View the contents of the CRL file
 - **openssl -crl -in /opt/tak/certs/files/ca.crl -inform PEM -text -noout**
- Search the list for a revoked certificate
 - **openssl -crl -in /opt/tak/certs/files/ca.crl -inform PEM -text -noout | grep <revokedSN>**
 - **Example:** `openssl -crl -in /opt/tak/certs/files/ca.crl -inform PEM -text -noout | grep 5D7DE36856929631`

```
bash-4.4$ openssl crl -in /opt/tak/certs/files/ca.crl -inform PEM -text -noout
Certificate Revocation List (CRL):
  Version 1 (0x0)
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C = XX, ST = XX, L = XX, O = TEST, OU = TEST, CN = TAK-ID-CA-01
  Last Update: Aug  5 14:40:22 2024 GMT
  Next Update: Aug  5 14:40:22 2026 GMT
Revoked Certificates:
  Serial Number: 5D7DE36856929631
    Revocation Date: Aug  5 14:21:45 2024 GMT
  Serial Number: F67C7233F499BF1D
    Revocation Date: Aug  5 14:40:22 2024 GMT
  Signature Algorithm: sha256WithRSAEncryption
1b:41:88:e0:9b:71:0f:3f:66:e0:c2:d5:26:65:a9:b6:27:ff:
78:e7:5d:b8:6f:e9:83:b1:a3:65:4e:a0:fe:99:4e:e8:db:66:
99:79:a5:ca:17:0c:8b:d2:41:2b:fb:d1:49:26:18:93:5c:ed:
72:03:e0:b4:e7:fc:79:d4:46:7c:1e:e1:d0:1c:f5:04:f4:77:
b3:0d:4c:ea:b5:e7:04:f8:4f:a0:61:71:68:f7:5d:01:71:3e:
47:02:d3:ee:0f:0e:57:93:2b:82:98:4e:e8:00:f2:23:85:4c:
fb:80:11:17:77:6f:81:4c:48:d6:92:da:15:b2:56:8a:b8:9b:
fb:7d:aa:3b:d9:20:bc:9d:fd:c1:b6:84:2d:3c:92:62:ca:bf:
c5:60:37:1c:18:57:0c:70:db:13:f6:5b:b2:8f:2e:f4:69:28:
1e:f8:5a:35:24:9f:ad:4f:3b:25:43:c4:8a:1e:29:9c:06:91:
6b:26:3a:53:86:3c:4c:21:45:1b:3e:8d:02:c1:da:f0:87:18:
c9:06:23:be:2e:51:74:ce:96:54:fb:83:cc:56:35:89:10:05:
f2:ca:a3:b5:a6:f3:67:69:dc:00:77:3b:23:59:13:91:aa:45:
c2:74:64:4e:df:79:c6:2c:4d:b6:cb:b5:76:a9:4c:00:a0:60:
e9:36:30:c7
bash-4.4$
```

Revoking Clients - Publish the Revoked Certificate - Command Line

- Switch to the tak user
 - **sudo su tak**
- Change the directory to the tak directory
 - **cd /opt/tak**
- Edit the CoreConfig file
 - **vi CoreConfig.xml**
 - Look for the following area of code towards the bottom of the file.

```
<security>
  <tls keystore="JKS" keystoreFile="certs/files/TAK-Server.jks" keystorePass="atakatak" truststore="JKS" truststoreFile="certs/files/truststore-TAK-ID-CA-01.jks" truststorePass="atakatak" context="TLSv1.2" keymanager="SunX509"/>
</security>
```


Revoking Clients - Publish the Revoked Certificate - Command Line

- At the end of the “tls” line remove the “/”.
 - On a new line add the following line of code:
 - **<crl_name=“TAKServer CA” crlFile=“certs/files/<your CA>.crl”/>**
 - **Example:** <crl_name=“TAKServer CA” crlFile=“certs/files/TAK-ID-CA-01.crl”/>
 - Add a closing statement for tls element on a new line.
 - **</tls>**
 - Save and exit the file.
 - press the “esc” key
 - type the following:
 - **:wq!**
 - press enter

```
</certificatesigning>
<security>
  <tls keystore="JKS" keystoreFile="certs/files/TAK-Server.jks" keystorePass="atakatak" truststore="JKS" truststoreFile="certs/files/truststore-TAK-ID-CA-01.jks" truststorePass="atakatak" context="TLSv1.2" keymanager="SunX509">
    <crl_name="TAKServer CA" crlFile="certs/files/TAK-ID-CA-01.crl"/>
  </tls>
</security>
```

Revoking Clients - Publish the Revoked Certificate - Command Line

- Validate the file.
 - **./validateConfig.sh**
- Once validated exit the tak user
 - **exit**
- Restart the server
 - **sudo systemctl restart takserver.service**
- You can monitor the server on restart to verify there are no errors:
 - **sudo tail -f /opt/tak/logs/takserver-messaging.log**

```
bash-4.4$ ./validateConfig.sh
CoreConfig.xml validates
bash-4.4$
```

Revoking Certificates - Command Line

- Determine the certificate issuing signing certificate authority
 - **openssl x509 -text -in /opt/tak/certs/files/<commonName>.pem | grep -Eo "Issuer:."**
 - **Example:** `openssl x509 -text -in /opt/tak/certs/files/revokedClient.pem | grep -Eo "Issuer:."`
- Revoke the Client:
 - Change to the tak user
 - **sudo su tak**
 - Change directories to the certificate directory
 - **cd /opt/tak/certs**
 - Revoke the targeted cert
 - **./revokeCert.sh /opt/tak/certs/files/<commonName> /opt/tak/certs/files/ca-do-not-share /opt/tak/certs/files/cat**
 - **Example:** `./revokeCert.sh /opt/tak/certs/files/revokedClient /opt/tak/certs/files/ca-do-not-share /opt/tak/certs/files/ca`

```
[takadmin@localhost logs]$ sudo su tak
bash-4.4$ cd /opt/tak/certs
bash-4.4$ ./revokeCert.sh /opt/tak/certs/files/player1 /opt/tak/certs/files/ca-do-not-share /opt/tak/certs/files/ca
Using configuration from ../config.cfg
Adding Entry with serial number F67C7233F499BF1D to DB for /C=XX/ST=XX/L=XX/O=TEST/OU=TEST/CN=player1
Revoking Certificate F67C7233F499BF1D.
Data Base Updated
Using configuration from ../config.cfg
```